

Podium: An Open-Source Library for Rendezvous and Docking Guidance, Navigation, and Control with Integrated Formal Verification

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Abstract

Podium is an open-source Python library for developing, testing, and formally validating guidance, navigation, and control (GNC) algorithms for spacecraft rendezvous, proximity operations, and docking (RPOD) in low and medium Earth orbit. It provides relative-motion models, convex and sequential-convex trajectory planners, a relative-navigation filter and controllers, a deterministic simulator, and a verification layer. A set of flight kernels—the relative-motion, navigation, and attitude routines—is written in a restricted, statically analyzable Python subset, translated to C, shown alarm-free on contracted (or explicitly assumed) input ranges by the Frama-C Evolved Value Analysis (EVA) plug-in, and compiled with the CompCert verified compiler, with the emitted C checked against the Python reference on golden vectors. The barrier, KKT, and optimality-gap guarantees are re-verified in exact rational arithmetic in the trusted checker: tolerance-free barrier certificates for abort safety, Karush–Kuhn–Tucker (KKT) certificates that, when a valid Lagrangian dual point exists, give an exact rational bound on the suboptimality of an approximate online convex solve, and tolerance-free bounds on the global optimum of a nonconvex keep-out quadratic program representative of the docking geometry. This paper describes the architecture, the verification approach, and a reproducible end-to-end example.

1 Motivation and significance

Spacecraft rendezvous and docking place strong demands on software [1]: the maneuvers are safety-critical, some unsafe commands or contact events cannot be corrected after the fact, and the flight code must be shown to be free of runtime errors and faithful to its specification. Mature simulation environments exist, including Basilisk [2], NASA’s 42 [3], Orekit [4], and the General Mission Analysis Tool (GMAT) [5], and Podium does not attempt to replace them for general astrodynamics. Its purpose is narrower: to organize development around spacecraft rendezvous, proximity operations, and docking (RPOD) rather than absolute orbit propagation, and to make formal validation part of ordinary development.

The library is motivated by three needs. First, relative motion, approach corridors, keep-out zones, plume impingement, and docking contact are central to RPOD and are a modeling focus distinct from absolute orbit propagation. Second, convex trajectory optimization is now standard in guidance [6, 7, 8, 9], but a floating-point quadratic program (QP) or second-order cone program

*Starcloud, Inc. Source and documentation: github.com/adi-oltean/podium. Archived at <https://doi.org/10.5281/zenodo.21225268>.

(SOCP) solver can return a nominally optimal solution that violates a constraint under rounding, so Podium re-checks such solver output with an exact-rational Karush–Kuhn–Tucker (KKT) checker that computes an exact rational bound on the solution’s suboptimality. Third, the gap between a Python prototype and analyzable flight-target C, ordinarily crossed by hand, is here crossed mechanically, so that validation evidence applies to the same code path intended for deployment.

2 Software description

2.1 Architecture

Figure 1 shows the two paths this description follows: the restricted Python-to-C flight-code path (top) and the exact-rational certificate layer (bottom). Podium is organized as subpackages under `src/podium`. The `core` package holds the verifiable static subset: the Clohessy–Wiltshire dynamics and state-transition matrix [10], the Yamanaka–Ankersen matrix for elliptic orbits [11], relative orbital elements with Keplerian, J_2 , and J_2 -plus-drag state-transition matrices [12], a quaternion kernel, and fixed-step integrators. The `dynamics` package holds the truth models: nonlinear relative motion with J_2 and differential drag over a seeded stochastic atmosphere, and rigid-body attitude with a suite of environmental torques (gravity gradient, aerodynamic, solar radiation pressure, and magnetic).

The `guidance` package implements convex and sequential-convex trajectory planners on the relative-motion discretizations. The `control` package provides discrete and continuous linear-quadratic regulators, quaternion attitude feedback, and a push-only thruster allocation. The `nav` package implements a relative-navigation extended Kalman filter with a Joseph-form (numerically stable) covariance update and sensor models for relative Global Navigation Satellite System (GNSS) navigation, camera, and lidar. The `sim` package provides a deterministic fixed-step engine with bit-identical replays, Signal Temporal Logic (STL) specification oracles, and contact checking. The `verify` and `emit` packages provide the verification layer and the C code generator described below.

2.2 The verifiable flight kernels

The flight kernels are written as pure step functions in a restricted subset of Python: statically shaped arrays, bounded loops, no dynamic allocation, and machine-readable contracts on input ranges and invariants. The subset is chosen so that a sound static analyzer can reason about it and so that its scalar and subscript statements translate line-for-line to C, with array expressions lowered to bounded row-major loops. The `emit` package generates C99 from the subset, annotated with ACSL (ANSI/ISO C Specification Language) contracts, together with correctly rounded implementations of sine and cosine from CORE-MATH [13]. The Frama-C/EVA abstract-interpretation analyzer [14] shows the emitted C free of runtime alarms (division by zero, overflow, invalid access) on the contracted input ranges. The generated C is checked against the Python reference on a suite of golden vectors: bit-exact for scalar arithmetic and square root, and, in the correctly rounded (CORE-MATH) mode, for sine and cosine as well; the remaining libm transcendentals (for example, the inverse-trigonometric calls in the anomaly kernels) stay within a one-unit-in-the-last-place tolerance, and matrix products agree only up to floating-point reassociation, since the emitted row-major loops and NumPy’s Basic Linear Algebra Subprograms (BLAS) accumulate in a different order. The vectors are regenerated deterministically from a fixed per-kernel seed rather than stored as fixtures, so the suite is reproduced identically on every run and cannot drift. The same vectors are replayed through the CompCert verified compiler [15], which machine-checks semantics preservation from C to assembly on x86-64 for the supported C subset and calling convention; the emitted C

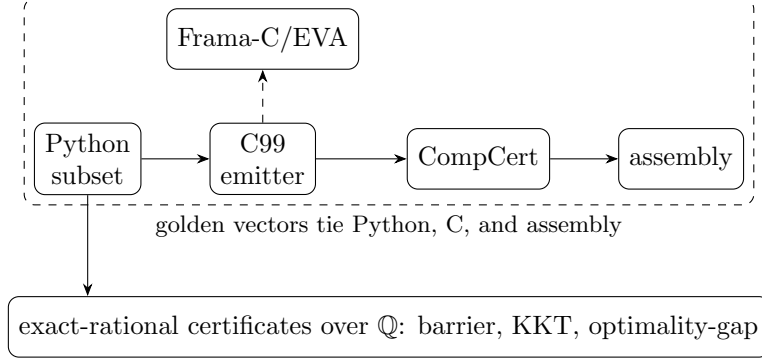


Figure 1: The flight-code path (top): a restricted Python subset is emitted to C99, shown alarm-free by Frama-C/EVA, and compiled by CompCert, with golden vectors tying the Python, C, and assembly on the tested inputs. A separate layer re-runs the exact-rational certificates (verified over $\mathbb{Q}$) together with reachability checks on the relevant path changes.

additionally reproduces the golden vectors bit-for-bit on a second instruction-set architecture (ISA), AArch64, under an ordinary cross-compiler and emulator; this cross-architecture bit-exactness is that of the scalar arithmetic/square-root class and relies on IEEE-754 binary64 with fused-multiply-add contraction disabled (`-ffp-contract=off`, SSE2 on x86-64), the libm and matrix-product classes remaining within the tolerances above. The golden vectors thus tie the Python reference, the emitted C, and the compiled assembly on the tested inputs. This emitted, analyzed, and compiled path covers the flight kernels—the relative-motion state transitions, the navigation-filter update, and the quaternion routines—rather than the guidance planners or controllers, whose online solves are instead covered by the certificate layer below.

Software metadata. Language Python 3.10+; license MIT; NumPy required, with optional CVXPY, Clarabel, and ECOS for the solver-backed paths and optional verification tooling (Frama-C, CompCert, JuliaReach). Continuous integration (CI) runs on GitHub Actions. The examples and the per-release audit bundle are the primary distribution channel; the library is in active development (v0.x).

3 Verification approach

Podium runs verification as a set of automated checks re-executed on every relevant change, rather than as a manual milestone; Table 1 summarizes them.

Two implementation choices support these guarantees. First, several certificates are checked in exact rational arithmetic over the rationals $\mathbb{Q}$, with no floating point in the trusted checker: a barrier for abort safety [18, 19], KKT conditions for the online convex solves, and bounds on the global optimum of a nonconvex keep-out QCQP—a primitive of the docking geometry, not yet wired into the full-horizon planner—through the S-procedure [20, 21] and its trust-region special case [22]. The barrier and optimality-gap certificates are tolerance-free, their acceptance being an exact positive semidefiniteness test together with exact feasibility; exact rational certificates of this kind build on a body of work in sum-of-squares and polynomial optimization and in verified semidefinite programming [23, 24, 25, 26, 27]. The same exact-arithmetic checker also discharges control-Lyapunov and sum-of-squares certificates, exercised in the test suite though not part of the reference mission; the constructions and their soundness proofs are collected in a companion technical

Table 1: Automated verification checks, re-run in continuous integration on the relevant path changes (and on scheduled drift checks where noted). QP: quadratic program. SOCP: second-order cone program. QCQP: quadratically constrained quadratic program. ISA: instruction-set architecture.

Check	Guarantee
Closed-loop reachability (JuliaReach [16])	Flowpipes avoid unsafe sets on the benchmark models [17]
Exact-rational barrier certificates	Infinite-horizon abort safety, verified over $\mathbb{Q}$
Exact-rational KKT certificates	Online QP solves given an exact rational suboptimality bound; SOCP solves re-verified against conic KKT; both over $\mathbb{Q}$
Exact optimality-gap certificates	Bounds on a nonconvex keep-out QCQP optimum, verified over $\mathbb{Q}$
Sound static analysis (Frama-C/EVA)	Emitted C alarm-free on contracted or assumed input ranges
Verified compiler and cross-ISA	Golden vectors preserved through CompCert and a second ISA
Independent physics oracle	Translational truth model cross-validated against Orekit [4]

note in the repository. The KKT check re-verifies an approximate solver’s output: it computes the optimality residuals exactly and, when a valid Lagrangian dual point exists, reports an exact rational bound on the point’s suboptimality. For a strictly convex quadratic program this bound accounts for the stationarity residual through the problem’s curvature; a small residual alone is not a suboptimality bound, since with a rank-deficient objective (a linear program or minimum-fuel problem) an approximately stationary point can be arbitrarily suboptimal. For a second-order cone program, whose objective is linear, a rigorous bound instead requires exact conic-dual feasibility. The numerical solvers and the rational recovery that propose each certificate are untrusted; the trusted checker rationalizes the candidate and either verifies it exactly or refuses, so solver rounding can force a refusal or a weaker bound but never a false certificate. In every case the checker itself introduces no rounding error. Second, the emitted flight-code C is tied to the Python reference by the golden vectors (above) and carried to assembly by CompCert, so the validated algorithm and the compiled algorithm agree on the tested inputs. Every tagged release ships a byte-deterministic audit bundle: a build script gates the mission capture, the temporal-logic and docking-interface margins, and the barrier certificate, and emits the bundle alongside the generated flight C, a fresh Frama-C/EVA report, and a SHA-256 manifest over the byte-deterministic files with the recorded toolchain versions, so that an identical-source rebuild is independently verifiable.

Figure 2 and Table 2 make these certificates concrete on the keep-out QCQP primitive. As the rational dual multiplier λ is rounded toward its optimum λ^* , the exactly certified lower bound $g(\lambda)$ approaches the global optimum J^* at the rate the S-procedure predicts: quadratically when the trust-region subproblem is nonsingular, and linearly in the hard case where $A(\lambda^*) = P_0 - \lambda^*P_1$ is rank-deficient. Every plotted value is an exact rational verified by the checker. The KKT check behaves analogously (Table 2b): it returns an exact rational suboptimality bound only when a valid dual point exists, and refuses otherwise. A tolerance-based test would accept the rank-deficient point—stationarity residual 10^{-10} , yet 100 from the optimum—as optimal; the exact checker reports no bound rather than a false certificate.

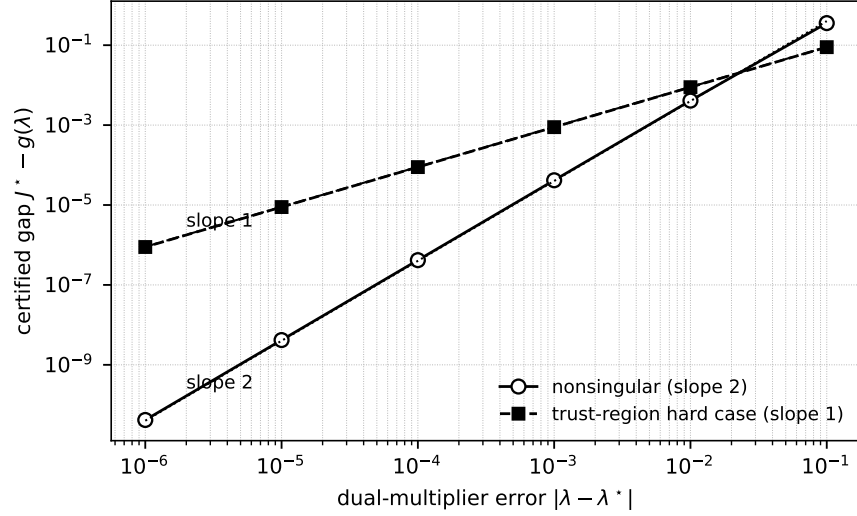


Figure 2: Exact optimality-gap recovery for a nonconvex keep-out QCQP. As the rational dual multiplier λ approaches its optimum λ^* , the gap between the certified lower bound and the global optimum, $J^* - g(\lambda)$, closes quadratically (slope 2) when $A(\lambda^*)$ is nonsingular and linearly (slope 1) in the trust-region hard case. Both coordinates are exact rationals verified over $\mathbb{Q}$; the guides mark slopes 2 and 1.

Table 2: Exact-rational certificate receipts, verified over $\mathbb{Q}$ by the trusted checker. (a) Optimality-gap recovery for the keep-out QCQP of Fig. 2. (b) The KKT check returns an exact suboptimality bound only when a valid dual point exists; exact residuals alone are not an optimality certificate.

<i>(a) Optimality-gap bracket recovery</i>			
Case	λ	$ \lambda - \lambda^* $	$J^* - g(\lambda)$
Nonsingular $\lambda^*=2/5, J^*=4$	39/100	1/100	1/244
	399/1000	1/1000	1/24040
	3999/10000	1/10000	1/2400400
Hard case $\lambda^*=1, J^*=-4/3$	101/100	1/100	803/90300
	1001/1000	1/1000	8003/9003000
<i>(b) KKT check: exact bound or refusal</i>			
Instance	Stationarity	Complementarity	Reported bound
PD QP $\frac{1}{2}x^2, x=1$	1	0	1/2 (certified)
Rank-deficient LP, $x=0$	10^{-10}	0	— (refused)
Indefinite $-\frac{1}{2}x^2, x=0$	0	0	— (refused)

4 Illustrative example

The reference mission (`podium.sim.mission`, exercised by `test_mission.py` in continuous integration) composes the layers end to end and is the paper’s worked example. A penalized trust-region (PTR) sequential-convex planner produces a closed-loop rendezvous approach from the barrier-certified start set to the docking target; the plan is propagated through the nonlinear relative-motion truth model, and the resulting closed-loop trajectory (Figure 3) is checked against range-channel Signal Temporal Logic phase specifications. The passive-drift abort safety of the start set (the safe free-drift trajectory after a thruster cutoff) is certified in exact rational arithmetic by the barrier—an invariant of the linear Clohessy–Wiltshire drift for a fixed safe formation, which bounds the linearized abort. Richer truth forces (J_2 , differential drag) and additional corridor/keep-out specifications are supported but are not enabled in this reference configuration. The exact-rational KKT and conic KKT checkers re-verify an online QP or SOCP solve against its own optimality conditions—for a second-order cone program, an embedded conic solver (ECOS) when it is installed. They run in continuous integration as part of the verification suite (`test_kkt.py`) rather than inside the reference-mission loop.

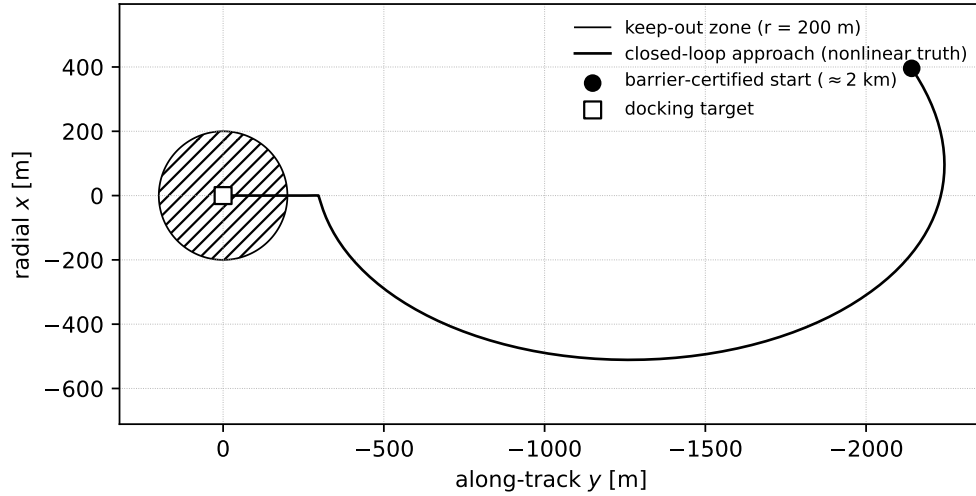


Figure 3: Reference mission: the closed-loop rendezvous approach from the barrier-certified start set (≈ 2 km) to the docking target under the nonlinear relative-motion truth model, in the local vertical/local horizontal (LVLH) frame. The transfer keeps clear of the keep-out zone and completes on the docking corridor; the passive-abort safety of the start set is certified in exact rational arithmetic by the barrier.

For a simpler entry point, the one-command demo `examples/vbar_approach.py` plans a shorter V-bar glideslope approach on the Clohessy–Wiltshire discretization and propagates it under linear Clohessy–Wiltshire dynamics (Python 3.10+, dependencies in `pyproject.toml`, no other setup); it is a readable on-ramp rather than the verified pipeline. An interactive web playback of that plan is linked from the repository, and the emitted flight kernels are additionally exercised in `examples/cfs_nav_app/`, a NASA Core Flight System (cFS) application, with build and run steps in its `README`.

Table 3: Feature coverage of representative open tools: each covers at most one or two of the three prongs Podium combines. ✓ full, ~ partial, blank none. Prong (i): a sequential convex programming (SCP) planner; (ii): exact-rational per-solve certificates re-run in continuous integration; (iii): a restricted high-level-source-to-C path checked by golden vectors, Frama-C/EVA, and CompCert. RPOD marks a rendezvous/docking focus.

	(i) SCP	(ii) cert/CI	(iii) verified C	RPOD
SCPToolbox.jl [28]	✓			~
GuSTO.jl [29]	✓			~
ct-scvx [30]	✓		~	~
successive_rendezvous [31]	✓			✓
RealCertify [32]		~		
ValidSDP [33]		~		
Verified SDP [34]		~		
Copilot [35]			✓	
Credible autocoding [36, 37]			~	
CVXPYgen [38]			~	
Basilisk [2]				✓
Podium	✓	✓	✓	✓

5 Impact and comparison

Each ingredient of Podium is well covered by open prior art; the contribution is their coupling. Table 3 makes this precise against the closest tools on each axis.

On trajectory optimization, the open SCPToolbox.jl [28] and GuSTO.jl [29] implement the sequential-convex family with rendezvous and free-flyer examples; ct-scvx [30] adds continuous-time feasibility with in-house C generation, and successive_rendezvous [31] is a released six-degree-of-freedom Apollo transposition-and-docking implementation. Prong (i) for RPOD is thus established open art, and none of these carries a certificate or verified-code layer. On certificates, RealCertify [32], the Coq-checked ValidSDP [33], and verified SDP [34] produce exact-rational or rigorously bounded certificates for general polynomial and conic problems, and certifiable trajectory optimization [39], certifiable perception [40], and exact SDP relaxations for QCQPs [41, 42] produce certified bounds by numerical semidefinite programming; but none is online, RPOD-specific, tolerance-free over $\mathbb{Q}$, and re-run in continuous integration. On verified code, the Copilot toolchain [35] couples a restricted high-level source with a C path, formal proof, and CI (for runtime monitors, not guidance); credible autocoding [36, 37] and verified model predictive control solvers [43] autocode convex-optimization solvers to contract-annotated C, and execution-time-certified embedded QP solvers [44] bound the online iteration count; verified spacecraft control programs [45] and verified optimization reductions in a proof assistant [46] certify code or problem transformations formally; and CVXPYgen [38] generates C from Python convex problems; none couples exact-rational per-solve certificates with an RPOD library. The simulators Basilisk [2] and NASA’s 42 [3] and the flight-dynamics systems Orekit [4] and GMAT [5] support proximity-operations modeling but no planner, certificate, or verified-code layer.

The novelty is therefore not any single prong but the integration of all three: a publicly released, RPOD-focused library that, in one codebase, couples sequential-convex planners with exact-rational *per-solve* certificates re-run in continuous integration and a restricted Python-to-C flight-code path checked by golden vectors, Frama-C/EVA under stated input contracts, and CompCert over its

supported subset. Podium is complementary to the tools above rather than a replacement. It is intended for researchers and engineers developing and validating RPOD guidance and control, and as a reproducible baseline: verification runs on every change and every release carries its evidence, so that results built on the library can be re-checked independently.

6 Limitations

The guarantees are deliberately scoped, and each is conditional on its assumptions: the sound-analysis result holds while inputs remain within the declared contracts, and the certificates hold for the stated models and problem data, so establishing at the system level that flight inputs stay within contract and that the models bound reality belongs to the surrounding assurance case rather than the library. The cross-architecture bit-exactness of the emitted C holds for the scalar arithmetic and square-root class, which is correctly rounded and hence bit-identical across conforming hardware under IEEE-754 binary64 with fused-multiply-add contraction disabled; correctly rounded sine and cosine are bit-exact in that mode, while the remaining libm transcendentals and matrix products fall in documented tolerance classes. The abort-safety barrier is certified exactly for the linear Clohessy–Wiltshire drift of a fixed formation—the standard e/i-vector passive-safety analysis, here machine-checked over $\mathbb{Q}$ —and bounds the linearized abort; the residual gap to the nonlinear, J_2 -and-drag, actuator- and sensor-perturbed trajectory is not certified by the barrier but is exercised by the nonlinear closed-loop reachability and temporal-logic checks, so exact synthesis on the linear model and high-fidelity nonlinear simulation are complementary rather than substitutes. The exact-rational checkers run offline in continuous integration as a pre-deployment gate that adds nothing to the flight loop; their rational entries stay small on the per-primitive problems, and bounding their growth at full trajectory scale is the open scaling direction. The verifiable subset is intentionally restricted to statically shaped arrays, bounded loops, and no dynamic allocation—the memory- and timing-determinism discipline of high-integrity flight software (MISRA-C, the JPL Power of Ten)—so the emitted kernels are allocation-free and of fixed, bounded complexity, and thus flight-suitable by construction; variable-horizon, adaptive, and event-triggered logic is deliberately excluded from this path and lives in the untrusted planner and controller layers, whose online solves the certificate layer covers. Finally, Podium produces verification *evidence*—sound-analysis and verified-compiler receipts, exact certificates, and golden-vector equivalence—of kinds recognized in high-assurance software practice, rather than a completed qualification under a standard such as DO-178C or ECSS; mapping that evidence onto a certification case, and benchmarking the generated kernels and solvers against production baselines, are outside the present scope.

7 Conclusions

Podium is an open library that organizes guidance, navigation, and control around RPOD scenarios from the outset and treats formal validation as a continuous activity. Its flight kernels, exact-arithmetic certificates, and verified flight-code path are designed so that validation evidence applies to the same code path intended for deployment, and so that its guarantees can be reproduced by any user. The positive semidefiniteness checks common to these certificates use an $O(n^3)$ symmetric $LDL^\top$ factorization, whose factors are the nonnegativity witness. Scaling the certificates from the per-primitive matrices to full trajectory horizons, and broadening mission coverage, are the main directions of continued development.

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